STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 12B

Description

12.Write R program to do the following tests with the sample data and visualize the results graphically.

b) t-test

Procedure and Overvation:

DESCRIPTION:

The T-test is one of the most common test which is used to determine whether the mean of the 2 groups is equal to each other the assumption
for the test is that both groups are sampled from a normal distribution with equal fluctuation the null hypothesis is that the 2 means are same and the alternative is that they are not identical.

->we can compute as T-test statistic that will follow at distribution with n1+n2-2-degreee of freedom

1.one sample T-test-one sample T-test is a T-test which compares the mean of a vector against theoretical theme there is a following formula which is used to compute the T-test

t=m-mu/t/root(n)

->where m-mean
s-standard deviation
n-is a no.of observations

SOURCE CODE:

```
set.seed(0)
ship_vol <- c(rnorm(70, mean = 35000, sd = 2000))
t.test(ship_vol, mu = 35000)
```

SOURCE CODE:

```
x  <- c(0.593, 0.142, 0.329, 0.691, 0.231, 0.793, 0.519, 0.392, 0.418)

t.test(x, alternative="greater", mu=0.3)
```

->independent sample T-test is equality and there variance the independent sample T-test variance one of the 3forms(one factor)

->independent sample T-test

y1&y2 are numeric values

->independent sample T-test where y1 is numeric ,y2 is binary

independent sample T-test variances not assumed.

pair sample T-tesst

1.2-samples

2.n1=n2

n1!n2

apply for sample T-test

->to perform paired sample T-test we need 2 vectors data y1&y2

T-test (y1,y2,paired=TRUE)

#create vectors to hold plant heights from each sample

SOURCE CODE:

```
#create vectors to hold plant heights for each sample
group1<-c(8,8,9,9,9,11,12,13,13,14,15,19)
group2<-c(11,12,13,13,14,14,14,15,16,18,18,19)
t.test(group1,group2,var.equal=TRUE)
```

```
        One Sample t-test

data:  ship_vol
t = -0.27702, df = 69, p-value = 0.7826
alternative hypothesis: true mean is not equal to 35000
95 percent confidence interval:
 34498.18 35379.45
sample estimates:
mean of x
 34938.81
```

```
        One Sample t-test

data:  x
t = 2.2051, df = 8, p-value = 0.0397
alternative hypothesis: true mean is greater than 0.3
95 percent confidence interval:
 0.3241111       Inf
sample estimates:
mean of x
0.4544444
```

```
        Two Sample t-test

data:  group1 and group2
t = -2.5305, df = 22, p-value = 0.01823
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -5.5904820 -0.5761847
sample estimates:
mean of x mean of y
 11.66667  14.75000
```

VIVA RESPONSES

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